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$2(3z^2 - z + 1) + i(z + 1) = 0$

$2(3z^2 - z + 1) + i(z + 1) = 0$

$6z^2 - 2z + 2 + iz + i = 0$

$6z^2 + (-2 + i)z + (2 + i) = 0$

Discrimant $D = (-2 + i)^2 - 4 \cdot 6 \cdot (2 + i)$

$= 4 - 1 - 4i - 48 - 24i$

$= -45 - 28i$

$$\begin{cases} x^2 - y^2 = -45 \\ 2xy = -28 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = -45 \\ x^2(-y^2) = -196 \end{cases}$$

De resolvente vierkantsvergelijking:

$\lambda^2 + 45\lambda + 196 = 0$

Discrimant $D = 45^2 - 4 \cdot (-196) = 2809$

$\lambda_{1,2} = \dfrac{-45 \pm 53}{2}$

$x = \pm 2$ en $y = \pm 7$

$b < 0$:

$D = (2 - 7i)$ of $D = (-2 + 7i)$

Oplossingen voor z:

$z_{1,2} = \dfrac{(2 - i \pm +(2 - 7i)}{12}$

$z_1 = \dfrac{1 - 2i}{3}$

$z_2 = \dfrac{i}{2}$

References